

TraceMail AI

Email Forensic Investigation Report

Case ID: TM-73CE47E4

Generated: 2026-09-06T01:46:01.732791

1. Executive Summary

Severity	CRITICAL
Risk Score	100/100
Classification	Business Email Compromise / Financial Phishing

Assessment: The email presents a high-confidence set of suspicious indicators and should be treated as a potentially malicious communication. Further investigation and containment are recommended.

2. Email Evidence

Filename	N/A
Sender	finance@paypa1-security.com
Recipient	employee@company.com
Reply-To	urgent-payment@protonmail.com
Subject	URGENT: Invoice Payment Required Today
Date	Sat, 05 Sep 2026 10:30:00 +0530
Message-ID	<123456789@example.com>

3. Email Authentication Findings

Authentication Results	mail.company.com; spf=fail; dkim=fail; dmarc=fail
Received-SPF	fail
DKIM Signature	False

4. Header & Relay Forensics

Total Relay Hops: 1

Earliest Observed Public IP: 8.8.8.8

Observed Public IPs:

- 8.8.8.8

The earliest observed public IP represents observed source infrastructure and should not automatically be treated as the attacker's physical location.

5. IP Intelligence

IP	Country	City	ISP	Risk
8.8.8.8	United States	Mountain View	Google	LOW

6. URL & Domain Intelligence

Extracted URLs: 1

- <http://paypa1-security.com/invoice/84721>

Extracted Domains: paypa1-security.com

7. Threat DNA

Feature	Value
spf	FAIL
dkim	FAIL
dmARC	FAIL
url	True
url_count	1
financial	True
urgency	True
reply_to_mismatch	True
attachment	False
hop_count	1
typosquatting	True

DNA Fingerprint:
SPF-FAIL|DKIM-FAIL|DMARC-FAIL|URL-1|FIN-1|URG-1|RPLY-1|ATT-0|HOP-1|TYPO-1

Risk Pattern: Business Email Compromise / Financial Phishing

8. Identified Risk Factors

- {'type': 'authentication', 'severity': 'high', 'message': 'SPF authentication failed'}
- {'type': 'authentication', 'severity': 'medium', 'message': 'DKIM authentication is missing or failed'}
- {'type': 'authentication', 'severity': 'high', 'message': 'DMARC authentication failed'}
- {'type': 'header', 'severity': 'high', 'message': 'Reply-To domain differs from sender domain'}
- {'type': 'content', 'severity': 'medium', 'message': 'Urgency language detected', 'matches': ['urgent', 'immediately', 'action required', 'today']}
- {'type': 'content', 'severity': 'high', 'message': 'Financial/payment-related intent detected', 'matches': ['payment', 'invoice', 'account']}
- {'type': 'url', 'severity': 'medium', 'message': '1 URL(s) detected in email'}

9. Evidence Integrity

Hash Algorithm: SHA-256

Evidence SHA-256:
59f1dc7bcd5926080987f1d49f631903072c96b47ff0b1fb6b92c4ae3c40ee61

SHA-256 hash uniquely identifies the analyzed email evidence and supports tamper-evident integrity verification.

10. Investigator Conclusion

The email presents a high-confidence set of suspicious indicators and should be treated as a potentially malicious communication. Further investigation and containment are recommended.

Report generated by TraceMail AI
Email Threat Detection & Forensic Intelligence Platform